A Formalization Program for Field 01

Phase Nodes, Scalar Profiles, Normal-Retention Interpretation, Memory, and Boundary Recording

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Abstract

This paper outlines a cautious formalization program for Field 01. It organizes the model's current mathematical development into a chain of concepts: phase, closed circulation, scalar modulus / VEV-profile layer, labelled normal-retention interpretation, gauge-like compensation, current-like flow, memory as an equivalence class, boundary recording, and reduced external description. The proposal is not presented as a completed physical theory, nor as a replacement for the Standard Model, quantum field theory, general relativity, or black-hole thermodynamics. Instead, it gives a structured toy-level framework in which the conceptual vocabulary of Field 01 can be compared with known mathematical structures, especially phase/gauge/vortex models. The central result is not a new fundamental equation, but a controlled formal vocabulary. The diagnostic layer translates selected Field 01 phrases into standard Abelian-Higgs benchmarks and records the toroidal self-confined-photon electron model of Williamson and van der Mark as a close conceptual precedent. The compatibility discussion further distinguishes the sectors: it shows that the scalar phase at zero modulus cannot represent a propagating photon, that a broken local $U(1)$ vortex has no independent massless phase, and that the two-gauge toy extension considered here places a massless mode and the vortex in orthogonal sectors. Within that restricted field content, Abelian kinetic mixing does not supply an open-to-closed transition. The horizon interpretation remains a bulk-to-boundary memory map and a reduced-access hypothesis. All claims remain provisional.

Keywords: Field 01; formalization program; open mode; closed sector; phase circulation; scalar modulus; normal-retention interpretation; Abelian-Higgs diagnostics; gauge mixing; toroidal photon model; memory; boundary recording; reduced density matrix.

1 How to Read This Paper

This paper is a formalization roadmap, not a completed theory. It should be read as the third research text in the current Field 01 sequence. The first paper introduced the elementary particle as a closed phase configuration. The second paper interpreted a black-hole horizon as a phase recording surface. The present paper asks a narrower question: what mathematical skeleton is needed for these claims to become discussable and testable?

Three levels must be kept separate:

1. **Established mathematics and physics**: phase fields, winding numbers, covariant derivatives, currents, reduced density matrices, vortex-like structures, and black-hole thermodynamic formulas.

2. **Field 01 interpretation**: phase as rhythm, a scalar modulus / VEV-profile layer as the carrier of a labelled normal-retention reading, memory as preserved phase structure, and horizon recording as a boundary representation of local phase data.
3. **Open hypotheses**: the identification of the scalar-profile layer, under the normal-retention interpretation, with particle-like mass; the use of boundary memory for horizons; and any relation to charge, spin, or black-hole information.

The aim is not to claim novelty where standard mathematics already exists. In fact, one important conclusion is that the first toy formalization of Field 01 naturally approaches known Abelian-Higgs or vortex-like structures. The possible value of Field 01 lies in its interpretation and in the proposed link between scalar-profile retention, memory, and boundary recording.

2 Standard-First Vocabulary Rule

This text follows a standard-first rule. Each technical object should be introduced first with the nearest standard vocabulary, then with the Field 01 reading, and finally with an explicit statement of what remains hypothetical. The reference map for this rule is maintained separately in `FIELD01_REFERENCE_MAP.md`.

- **phase rhythm**: first write phase variable, angular variable, or phase of a complex scalar; the rhythm language is Field 01-specific.
- **closed phase node**: first write vortex-like winding, topological defect, or winding number; the particle interpretation remains open.
- **scalar modulus / VEV-profile layer**: first write scalar amplitude, radial profile, or order-parameter modulus; the normal-retention interpretation is only the Field 01 label placed on this layer.
- **gauge-like compensation**: first write connection-like field, covariant derivative, or screening; this is not a derivation of electromagnetism.
- **memory**: first write preserved invariant data or equivalence class; the memory interpretation remains hypothetical.
- **boundary recording**: first write boundary data, reduced state, or holographic-style boundary language; no solution of black-hole information is claimed.

3 Formalization Chain

The current formalization can be summarized as

$$\begin{aligned} &\text{phase} \to \text{closed node} \to \text{scalar modulus / VEV-profile layer} \\ &\to \text{normal-retention interpretation} \to \text{gauge-like compensation} \to \text{current} \\ &\to \text{memory class} \to \text{boundary map} \to \text{reduced state}. \end{aligned} \tag{1}$$

This chain is not a derivation of known physics. It is a formalization route from the conceptual language of the model to mathematical objects that can be analyzed.

The main variables are

$$\mathcal{F}_{01}, \qquad \varphi, \qquad N \text{ or } \mathcal{N}, \qquad A_\mu, \qquad \mathcal{M}. \tag{2}$$

Here φ denotes a phase variable, N is the scalar toy profile or order-parameter modulus, $\mathcal{N}$ denotes the optional Field 01 normal-retention reading of that scalar layer, A_μ is a compensating connection-like field, and $\mathcal{M}$ denotes preserved phase-structural data. At this stage $\mathcal{F}_{01}$ is not yet defined by a final Lagrangian.

4 Phase And Closed Circulation

Let

$$\varphi : X \to S^1 \tag{3}$$

be a phase variable on an effective domain X. For a contour $\mathcal{C}$, define the phase circulation

$$\Gamma[\mathcal{C}] = \oint_{\mathcal{C}} d\varphi. \tag{4}$$

A closed phase node is represented by the condition

$$\oint_{\mathcal{C}} d\varphi = 2\pi n, \qquad n \in \mathbb{Z}. \tag{5}$$

Equivalently,

$$Q_{\text{wind}} = \frac{1}{2\pi} \oint_{\mathcal{C}} d\varphi = n. \tag{6}$$

In the toy model one uses the ansatz

$$\varphi_n(r, \theta) = n\theta. \tag{7}$$

This gives the desired winding, but a bare winding has long-range energy cost. On an annulus $a \le r \le R$,

$$\int_0^{2\pi} \int_a^R \frac{n^2}{r^2} r \, dr \, d\theta = 2\pi n^2 \log \frac{R}{a}. \tag{8}$$

Thus a bare phase winding is not yet a localized finite-energy particle-like object.

5 Scalar Profile, Normal-Retention Interpretation, And Mass Proxy

In vortex-like and Abelian-Higgs-type models, a winding configuration is usually accompanied by a scalar amplitude or radial profile. Field 01 first uses the same kind of scalar profile as a toy proxy. Only after this standard scalar layer is introduced does the Field 01 interpretation label it as normal retention:

$$N = N(r). \tag{9}$$

A minimal unscreened radial energy is

$$E_n[N] = 2\pi \int_0^R \left[\frac{A}{2}(N')^2 + \frac{B}{2}\frac{n^2 N^2}{r^2} + \frac{\lambda}{4}(N^2 - N_0^2)^2 \right] r\, dr. \tag{10}$$

The corresponding radial equation is

$$A\left(N'' + \frac{1}{r}N'\right) - B\frac{n^2}{r^2}N - \lambda N(N^2 - N_0^2) = 0. \tag{11}$$

In Field 01 language, the mass proxy is the energy of retaining a closed phase-scalar configuration, with the normal-retention interpretation added as a model-specific reading:

$$mc^2 \sim E_{\text{closed}}[\varphi, N]. \tag{12}$$

This is not yet a derivation of physical mass. It is only a formal expression of the idea that mass is associated with the energy cost of maintaining a closed local structure.

A numerical toy check with $A = B = \lambda = N_0 = 1$, $n = 1$, and $r \in [10^{-3}, 12]$ produced a particle-like profile with finite-disk energy approximately

$$E_{\text{unscreened}} \approx 9.0093. \tag{13}$$

A horizon-like boundary condition $N(R_H) = 0$ also admits a profile while preserving winding $Q_{\text{wind}} = 1$.

6 Connection-Like Compensation

In standard gauge notation, a connection field can screen or compensate a long-range phase gradient through a covariant derivative. The Field 01 toy model uses the same structure to represent compensation of retained phase mismatch. Introduce a connection-like field A_μ and the covariant phase derivative

$$D_\mu \varphi = \partial_\mu \varphi - A_\mu. \tag{14}$$

Under the gauge-like transformation

$$\varphi \mapsto \varphi + \chi, \qquad A_\mu \mapsto A_\mu + \partial_\mu \chi, \tag{15}$$

we have

$$D_\mu \varphi \mapsto D_\mu \varphi. \tag{16}$$

The field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{17}$$

is also invariant.

For the radial ansatz, the replacement is

$$\frac{n}{r} \longrightarrow \frac{n - a(r)}{r}. \tag{18}$$

A screened toy energy is

$$E_n[N,a] = 2\pi \int_0^R \left[\frac{A}{2}(N')^2 + \frac{B}{2}\frac{(n-a)^2 N^2}{r^2} + \frac{C}{2}\frac{(a')^2}{r^2} + \frac{\lambda}{4}(N^2 - N_0^2)^2 \right] r\, dr. \tag{19}$$

The radial equations are

$$A\left(N'' + \frac{1}{r}N'\right) - B\frac{(n-a)^2}{r^2}N - \lambda N(N^2 - N_0^2) = 0, \tag{20}$$

$$a'' - \frac{1}{r}a' + \frac{B}{C}(n-a)N^2 = 0. \tag{21}$$

For the same finite disk, the screened particle-like numerical profile gave

$$E_{\text{screened}} \approx 3.6340, \tag{22}$$

which is lower than the unscreened toy energy. This supports the mathematical role of compensation, but it does not yet derive electromagnetism.

7 Current-Like Object

A covariant-looking toy density may be written as

$$\mathcal{L}_{\text{toy}} = \frac{A_N}{2}(\partial_\mu N)(\partial^\mu N) + \frac{B}{2}N^2(D_\mu\varphi)(D^\mu\varphi) - V(N) - \frac{C}{4}F_{\mu\nu}F^{\mu\nu}. \tag{23}$$

The global phase-shift symmetry

$$\varphi \mapsto \varphi + \epsilon, \qquad \epsilon = \text{const} \tag{24}$$

gives the current-like object

$$J^\mu = \frac{\partial \mathcal{L}_{\text{toy}}}{\partial(\partial_\mu\varphi)} = BN^2 D^\mu\varphi. \tag{25}$$

The phase equation gives

$$\partial_\mu J^\mu = 0. \tag{26}$$

This is a conserved phase-compensation current inside the toy model. It is not yet electric current. It only shows that retained covariant phase flow has a conserved current-like expression.

8 Relation To Abelian-Higgs And Vortex Models

The screened toy model is close to known Abelian-Higgs or vortex-like structures. This must be stated explicitly. If one writes

$$\Psi = Ne^{i\varphi}, \tag{27}$$

then

$$|D_\mu\Psi|^2 = (\partial_\mu N)(\partial^\mu N) + N^2(\partial_\mu\varphi - A_\mu)(\partial^\mu\varphi - A^\mu). \tag{28}$$

Thus the pair (N, φ) is mathematically close to polar variables of a complex scalar field. Field 01 should not claim this structure as a new invention.

The possible contribution of Field 01 is interpretive:

- φ is read as phase rhythm or mode-1 circulation;
- N is first the scalar modulus / VEV-profile layer, then read in Field 01 through the normal-retention interpretation or local-depth label;
- winding is read as phase memory;
- A_μ is read as compensating field response;
- a localized gauge-invariant deformation of the scalar/gauge profile is read as a candidate normal-retention diagnostic.

Whether this interpretation leads to new mathematics or physics remains an open question.

9 Relation To Toroidal-Photon Electron Models

A related historical proposal is the toroidal-photon electron model of Williamson and van der Mark (1997). Their model and Field 01 share a conceptual aim: replace a material point core by a finite dynamical field configuration and associate rest energy with retained circulation.

This conceptual overlap does not make the two constructions mathematically identical. In the toroidal-photon model, closure refers to a framed electromagnetic energy-flow trajectory embedded in three-dimensional space. In the current Field 01 standard layer, closure is represented by the winding map

$$S^1_{\text{space}} \longrightarrow S^1_{\text{phase}}, \qquad Q_{\text{wind}} \in \mathbb{Z}, \tag{29}$$

with scalar and gauge profiles of an Abelian-Higgs-like radial vortex. Such a winding does not by itself produce a toroidal photon path, a localized three-dimensional particle, a Dirac spinor, or fermionic statistics.

The comparison is conceptual rather than a validation of either model. Results about charge, spin, magnetic moment, or confinement in a different ansatz cannot be transferred to Field 01 without a derivation in its own specified dynamics.

For the present program, this paper serves two purposes. It sets a novelty boundary: the general proposal of an electron or particle as a closed or toroidal electromagnetic wave has prior art. It also sharpens the open task: Field 01 must derive any three-dimensional localization, charge, spin, and observable consequences from its own defined dynamics rather than from visual similarity to a toroidal model.

10 Open And Closed Sector Compatibility

The Field 01 interpretation distinguishes an open mode 1 from a closed retained mode $(0 + 1)$. The present scalar/gauge benchmark does not yet realize that distinction physically. In the polar decomposition

$$\Psi = Ne^{i\varphi}, \tag{30}$$

the phase is undefined at $N = 0$, and its kinetic contribution is multiplied by N^2. Thus the schematic identification "photon = $N = 0$ open phase" is not a solution of the current model.

The minimal global and local candidates realize complementary parts of the desired structure. A globally broken $U(1)$ has a physical massless scalar phase mode, but an isolated two-dimensional vortex has logarithmically growing infrared energy. A locally broken $U(1)$ has a finite-energy Nielsen–Olesen vortex, but the phase is not an independent massless excitation; it supplies the longitudinal component of a massive vector. An unbroken local $U(1)$ keeps a massless gauge field but has no nontrivial scalar vacuum circle for the standard vortex mechanism.

A simple coexistence extension uses one complex scalar charged under two Abelian gauge directions, with canonical gauge kinetic terms and no other charged matter. In the scalar normalization used for the following matrix, its gauge mass matrix is

$$M^2 = N_0^2 \begin{pmatrix} e_A^2 & e_A e_B \\ e_A e_B & e_B^2 \end{pmatrix}, \qquad \mathrm{spec}(M^2) = \left\{0, N_0^2(e_A^2 + e_B^2)\right\}. \tag{31}$$

One orthogonal gauge combination remains massless, while the other is Higgsed and reduces to the standard finite-energy vortex construction. This permits coexistence in the toy construction, not conversion: the massless combination is absent from the scalar covariant derivative and does not close into the vortex.

Within this restricted field content, adding only Abelian kinetic mixing does not solve this transition problem. For $|\chi| < 1$, the term $-(\chi/2)F_Z^{\mu\nu}F_{\mu\nu}^P$ is removed by an exact field redefinition, leaving a free massless eigenmode and a rescaled massive vortex sector. Any induced flux in the original field basis is therefore not a trapped physical photon. A genuine open-to-closed mechanism would require a non-removable interaction and a new masslessness, topology, and finite-energy analysis.

For the current checkpoint, the Abelian-Higgs construction must be described as a closed-sector benchmark only. The physical representative of open mode 1 remains an open problem. Likewise, mode 0 should not be identified with the bare scalar value N; a safer diagnostic is a localized gauge-invariant departure from the vacuum profile, for example

$$\mathcal{R}_0(x) = \frac{\left(N^2(x) - N_0^2\right)^2}{N_0^4}. \tag{32}$$

This is a diagnostic convention, not a fundamental definition of fixation or memory.

11 Memory As Equivalence Class

Equivalence classes and invariant data are standard mathematical tools for separating physical or structural information from choices of representation. The Field 01 concept of memory is therefore formalized here as an equivalence class of preserved phase-scalar-gauge data, with normal retention kept as an interpretation of the scalar component:

$$\mathcal{M}_{\mathrm{bulk}} = [\varphi, N, A_\mu]_{\sim_{\mathrm{bulk}}}. \tag{33}$$

A first invariant set is

$$\mathcal{I}_{\mathrm{bulk}} = (Q_{\mathrm{wind}}, \Phi_F, \mathcal{J}, E_{\mathrm{class}}, \mathcal{B}), \tag{34}$$

where Q_{wind} is winding, Φ_F denotes flux or connection data, $\mathcal{J}$ denotes current flux, E_{class} is an energy class, and $\mathcal{B}$ denotes boundary-accessible phase data.

The equivalence relation identifies configurations that preserve these invariants while quotienting out gauge choice, coordinate representation, and small smooth deformations that do not change the record.

Boundary memory is written as

$$\mathcal{M}_\partial = [\varphi_\partial, A_\partial, F_\partial, Q_\partial]_{\sim_\partial}. \tag{35}$$

The bulk-to-boundary memory map is

$$\Pi_\partial : \mathcal{M}_{\text{bulk}} \to \mathcal{M}_\partial. \tag{36}$$

This is the formal version of the Field 01 statement that local memory becomes boundary phase record.

12 Boundary Reduction And Thermality

A standard reduced density matrix is obtained by tracing inaccessible degrees of freedom:

$$\rho_{\text{external}} = \text{Tr}_{\text{hidden}} \rho_{\text{full}}. \tag{37}$$

In Field 01 language, boundary memory may be split as

$$\mathcal{M}_\partial = (\mathcal{M}_{\text{acc}}, \mathcal{M}_{\text{hid}}, \mathcal{C}_{\text{corr}}), \tag{38}$$

where $\mathcal{M}_{\text{acc}}$ is accessible boundary data, $\mathcal{M}_{\text{hid}}$ is inaccessible data, and $\mathcal{C}_{\text{corr}}$ denotes correlations.

Then one may write schematically

$$\rho_{\text{external}} = \text{Tr}_{\mathcal{M}_{\text{hid}}} \rho(\mathcal{M}_\partial). \tag{39}$$

A reduced state may appear thermal,

$$\rho_{\text{external}} \approx \frac{e^{-\beta H_{\text{eff}}}}{Z}. \tag{40}$$

The Field 01 interpretation is that thermality may reflect reduced access to the full boundary record, not necessarily destruction of the record. This does not derive Hawking radiation or solve the black-hole information problem. It is only a formal expression of the model's interpretive stance.

13 Technical Appendix: Standard Diagnostics For Current Terminology

This appendix records the current standard-first diagnostic layer. It does not claim a new particle model, a new force, a new topological invariant, or a solution of the black-hole information problem. Its narrower purpose is to state which Field 01 terms have been translated into standard Abelian-Higgs-type calculations or into explicit relation-class bookkeeping. All Field 01 language in this section remains interpretive unless it is tied to a standard calculated quantity.

13.1 Standard Vortex Substrate

Use a complex scalar field and Abelian gauge profile in polar variables,

$$\Psi = N(r)e^{in\theta}, \qquad A = a(r)\,d\theta. \tag{41}$$

The radial energy convention used for the diagnostic benchmark is

$$E = 2\pi \int_0^R dr \left[\frac{r}{2}(N')^2 + \frac{(n-a)^2 N^2}{2r} + \frac{(a')^2}{2g^2 r} + \frac{\lambda r}{4}(N^2 - N_0^2)^2 \right]. \tag{42}$$

The finite-energy boundary behavior is represented by

$$N(0) = 0, \qquad a(0) = 0, \qquad N(R) \approx N_0, \qquad a(R) \approx n. \tag{43}$$

At the critical point in this normalization,

$$\lambda_{\mathrm{BPS}} = \frac{g^2}{2}. \tag{44}$$

For $g = 1$, $N_0 = 1$, $n = 1$, and $\lambda = 0.5$, the numerical benchmark gives

$$E_{\mathrm{total}} = 3.141567, \qquad \frac{E_{\mathrm{total}}}{\pi} = 0.999992. \tag{45}$$

This is standard Abelian-Higgs vortex machinery. Field 01 does not claim this calculation as new. Its role here is to fix the standard substrate before adding any interpretation.

13.2 Scalar Boundary Diagnostic

To test the phrase "loss of retained profile," compare the standard outer behavior $N(R) = N_0$ with a forced outer-zero condition $N(R) = 0$, while keeping the gauge endpoint $a(R) = n$. Define

$$\Delta E_{\text{outer zero}} = E[N(R) = 0, a(R) = n] - E[N(R) = N_0, a(R) = n]. \tag{46}$$

The raw penalty grows approximately with circumference. The normalized penalty

$$\sigma_R = \frac{\Delta E_{\text{outer zero}}}{2\pi R} \tag{47}$$

approaches the standard scalar half-kink boundary tension. For $\lambda = 1$ and $N_0 = 1$,

$$\sigma_R \to \frac{2}{3\sqrt{2}} \approx 0.471404521. \tag{48}$$

More generally, the flat-boundary scaling is

$$\sigma_\infty(\lambda, N_0) = \frac{2}{3\sqrt{2}} \sqrt{\lambda}\, N_0^3. \tag{49}$$

Thus, in this diagnostic, the Field 01 phrase "loss of retained profile" reduces to standard scalar boundary-layer physics. This does not prove a new invariant, a particle mass, or a new boundary law.

13.3 Gauge-Compensation Diagnostics

The first gauge-compensation diagnostic keeps $N(R) = N_0$ and varies the endpoint gauge profile $a(R) = \alpha$. Define

$$\Delta E_a(R, \alpha) = E[N(R) = N_0, a(R) = \alpha] - E[N(R) = N_0, a(R) = n]. \tag{50}$$

For the finite-disk diagnostic, the leading behavior is

$$\Delta E_a(R, \alpha) \approx \frac{\pi}{gR}(n - \alpha)^2. \tag{51}$$

Endpoint failure of gauge compensation is therefore read here as a standard finite-boundary mismatch cost.

The second diagnostic keeps the endpoint flux fixed and perturbs only the interior profile,

$$a_\varepsilon(r) = a_0(r) + \varepsilon f(r), \qquad f(0) = f(R) = 0. \tag{52}$$

The leading cost is the standard second variation,

$$\Delta E[\varepsilon f] = \varepsilon^2 K[f] + O(\varepsilon^3), \tag{53}$$

with

$$K[f] = 2\pi \int_0^R dr \left[\frac{N^2 f^2}{2r} + \frac{(f')^2}{2g^2 r} \right]. \tag{54}$$

This identifies compensation stiffness with the positive Hessian / second variation around the standard vortex solution. It does not define a new compensation invariant or a new interaction law.

13.4 Two-Node Relation As Resolution-Class Bookkeeping

A two-node retained relation is currently represented as a resolution-dependent equivalence class, not as a force. A gauge-aware relation class may be written schematically as

$$I^R_{\rm rel}(A, B) = \left(W_{\rm pair}, D_{\rm overlap}(A, B), \beta_{\rm regime}, P^\gamma_q(A, B), B_{\rm pair}(A, B), \gamma_{\rm class}(A, B) \right). \tag{55}$$

Here $W_{\rm pair}$ is a winding-pair label, $D_{\rm overlap}$ is a distance/range bin motivated by profile overlap, $\beta_{\rm regime}$ is a standard type-I/BPS/type-II regime label, P^γ_q is a Wilson-line dressed phase bin, $B_{\rm pair}$ is a boundary-pair label, and $\gamma_{\rm class}$ records the connector or access-path class. In this reading, two configurations have the same retained relation only when the selected class data match under the declared resolution package R.

To reduce arbitrariness in the distance bins, define scalar and gauge tails

$$u_N(r) = 1 - N(r), \qquad u_a(r) = 1 - a(r), \tag{56}$$

and the combined normalized overlap proxy

$$O_{Na}(d) = \frac{\int d^2x \, [u^A_N u^B_N + u^A_a u^B_a]}{\int d^2x \, [u^2_N + u^2_a]}. \tag{57}$$

For the current standard profile, the combined-overlap thresholds are

$$O_{Na} = 0.10 \quad \Rightarrow \quad d = 5.400085, \qquad O_{Na} = 0.01 \quad \Rightarrow \quad d = 8.642013. \tag{58}$$

A simple mass-weighted tail proxy gives close thresholds,

$$J_{\mathrm{mw}} = 0.10 \quad \Rightarrow \quad d = 5.183276, \qquad J_{\mathrm{mw}} = 0.01 \quad \Rightarrow \quad d = 8.440450. \tag{59}$$

This makes the distance bins less arbitrary, but it is still not a true two-vortex interaction energy. The bins remain diagnostic conventions unless a physical observable and cutoff rule are justified.

13.5 Gauge-Aware Phase Relation

A raw phase bin based on $\theta_B - \theta_A$ is gauge-sensitive. The safer standard object is the Wilson-line dressed phase along a connector γ:

$$h_\gamma(A, B) = \left[(\theta_B - \theta_A) - \int_\gamma A \right] \bmod 2\pi. \tag{60}$$

The corresponding phase bin is

$$P_q^\gamma(A, B) = \left\lfloor \frac{q h_\gamma(A, B)}{2\pi} \right\rfloor. \tag{61}$$

Because an open Wilson-line phase depends on the chosen connector unless a path/access rule is specified, the relation class must include γ_{class} or an explicit boundary-selected connector. The Field 01 reading is therefore not a raw relative phase, but a gauge-aware holonomy-dressed phase class under a declared access rule. This is not entanglement, not a force law, and not a new invariant beyond standard gauge-theory bookkeeping.

13.6 Current Diagnostic Status

The current technical status is deliberately modest. Several Field 01 phrases now have standard mathematical diagnostics attached to them:

- closed field object $\rightarrow$ Abelian-Higgs radial vortex ansatz;
- retained local profile $\rightarrow$ scalar modulus $N(r)$;
- loss of retention $\rightarrow$ scalar boundary layer / half-kink tension;
- phase compensation $\rightarrow$ gauge profile $a(r)$ and covariant derivative;
- compensation stiffness $\rightarrow$ second variation around a vortex profile;
- retained relation $\rightarrow$ resolution-dependent equivalence class;
- distance relation $\rightarrow$ overlap-motivated class bin;
- phase relation $\rightarrow$ Wilson-line dressed holonomy bin.

This is progress in vocabulary control, not a claim of new physics. The next burden is to test whether any of these diagnostics leads to a genuinely new, physically distinguishable structure rather than to a restatement of known Abelian-Higgs, vortex, gauge, or boundary-layer objects.

14 Immediate Reference Anchors

The following standard literature should be used as comparison anchors. These references do not support the Field 01 interpretation by themselves; they identify nearby established structures that the text must not claim as new.

- Nielsen and Olesen, "Vortex-line models for dual strings," *Nuclear Physics B* 61, 45–61 (1973), for the classic scalar-field, winding, and gauge-vortex comparison.
- Abrikosov, "On the magnetic properties of superconductors of the second group," *Soviet Physics JETP* 5, 1174–1182 (1957), for the type-II superconductivity vortex analogy.
- Manton and Sutcliffe, *Topological Solitons*, Cambridge University Press, 2004, for standard soliton, vortex, topological-charge, and moduli-space vocabulary.
- Vilenkin and Shellard, *Cosmic Strings and Other Topological Defects*, Cambridge University Press, 1994, for topological-defect language and cosmological context.
- Goldstone, Salam, and Weinberg, "Broken Symmetries," *Physical Review* 127, 965 (1962), and Higgs, "Broken Symmetries and the Masses of Gauge Bosons," *Physical Review Letters* 13, 508–509 (1964), for the global/local symmetry-breaking distinction.
- Weinberg, "A Model of Leptons," *Physical Review Letters* 19, 1264–1266 (1967), for a historical example of an unbroken massless gauge combination after symmetry breaking.
- Holdom, "Two U(1)'s and Epsilon Charge Shifts," *Physics Letters B* 166, 196–198 (1986), for Abelian gauge kinetic mixing.
- Bekenstein, Hawking, Page, and modern black-hole-information reviews, for any claim involving horizon entropy, thermality, reduced external states, or boundary information.

A related speculative precedent, rather than a standard validation source, is J. G. Williamson and M. B. van der Mark, "Is the electron a photon with toroidal topology?," *Annales de la Fondation Louis de Broglie* 22(2), 133–160 (1997). It is relevant to the novelty boundary for closed-wave particle imagery, but its self-confined-photon ansatz is not the same mathematical construction as the radial vortex model used here.

15 What Is Claimed And What Remains Open

This paper claims only the following:

- Field 01 can be organized into a coherent toy formalization chain;
- closed phase nodes can be represented by winding;
- a scalar modulus / VEV-profile layer can be modeled by a scalar profile and then given the Field 01 normal-retention interpretation;
- gauge-like compensation reduces long-range phase mismatch;
- a conserved current-like object follows from phase-shift symmetry;

- memory can be formalized as an equivalence class;
- boundary recording can be represented by a bulk-to-boundary map;
- reduced external descriptions can be interpreted as limited access to boundary memory.
- selected Field 01 terms can be tested against standard Abelian-Higgs, scalar-boundary, gauge-stiffness, overlap-bin, and Wilson-line diagnostics.
- the broad closed-wave or toroidal-field image has historical speculative precedents and is not claimed as a Field 01 invention.
- the present Abelian-Higgs construction is a closed-sector benchmark and does not derive the physical photon/open sector.
- a minimal two-gauge extension can make a massless spin-1 mode coexist with a finite-energy vortex, but neither orthogonality nor kinetic mixing supplies an open-to-closed transition.

The following remain open:

- a precise fundamental action;
- relation to the Standard Model;
- derivation of spin, charge, and particle spectrum;
- relation to the Higgs mechanism;
- comparison with Abelian-Higgs and vortex literature;
- construction of a boundary Hilbert space;
- derivation of Hawking radiation or black-hole entropy;
- testable physical predictions.
- a controlled two-vortex interaction-energy comparison;
- a canonical boundary/access rule for the connector class γ_{class}.
- any derivation identifying the closed node with a literal self-confined photon or a stable three-dimensional toroidal electromagnetic solution.
- a gauge-invariant physical representative of mode 1 that reproduces photon properties;
- a fundamental definition of mode 0 beyond a localized profile diagnostic;
- a non-removable interaction that couples a massless open eigenmode to the closed sector without destroying masslessness or finite energy.

16 Conclusion

The current mathematical development of Field 01 does not yet produce a completed physical theory. It does, however, produce a structured formalization path. Starting from phase circulation, one can define closed nodes, introduce a scalar modulus / VEV-profile layer, add the Field 01 normal-retention interpretation, add gauge-like compensation, obtain a current-like object, define memory as an equivalence class, map bulk memory to boundary memory, and interpret reduced external states as limited access to that boundary record. The diagnostic appendix adds a further discipline: key Field 01 phrases are tested against standard Abelian-Higgs benchmarks, scalar boundary-layer tension, gauge-compensation stiffness, overlap-motivated relation bins, and Wilson-line dressed phase bookkeeping.

The most important lesson is caution. Much of the mathematics resembles known vortex or Abelian-Higgs-like structures, and the broad closed-wave particle image also has earlier speculative precedents. The open/closed compatibility test now sharpens that boundary: the current vortex mathematics describes a closed sector, not a photon, while the two-gauge extension considered here leaves the open and closed sectors dynamically orthogonal. Field 01 must therefore be judged by whether it can define a physical open mode, a gauge-invariant localization structure, and a non-removable transition mechanism rather than by the presence of winding, toroidal imagery, or gauge-like notation alone.